



Arduino 4-Bit Binary Adder Project

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Understanding the Software

This project demonstrates a 4-bit binary adder implemented with an Arduino.

The system receives two 4-bit binary inputs, performs the addition using digital logic, and outputs the result along with the carry bit when the sum exceeds 4 bits.

The main goal is to apply digital electronics concepts such as binary addition and logic gates in a practical project.

Inputs and Outputs

- Pins 0–3: First 4-bit number
(Input A)
- Pins 4–7: Second 4-bit number
(Input B)
- Pins 8–11: 4-bit sum output
- Pin 12: Carry bit output

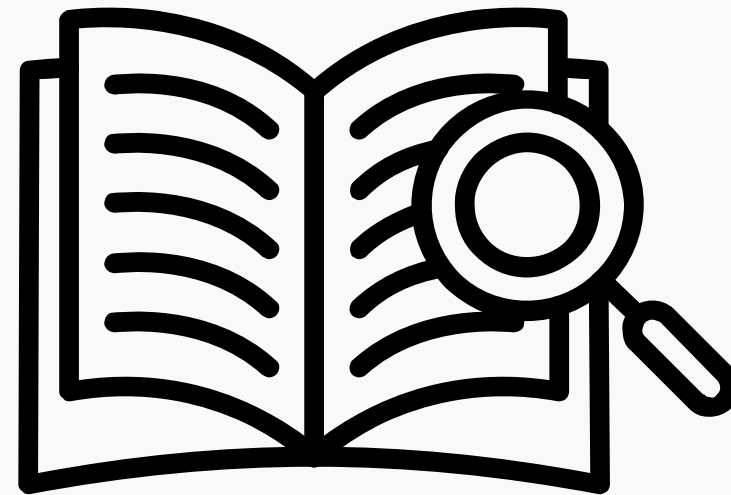
Developed Features

- Binary addition of two 4-bit numbers
- Real-time visualization through LEDs or the serial monitor
- Carry detection when the result exceeds 4 bits
- Structured Arduino sketch for continuous operation

Examples

Example 1

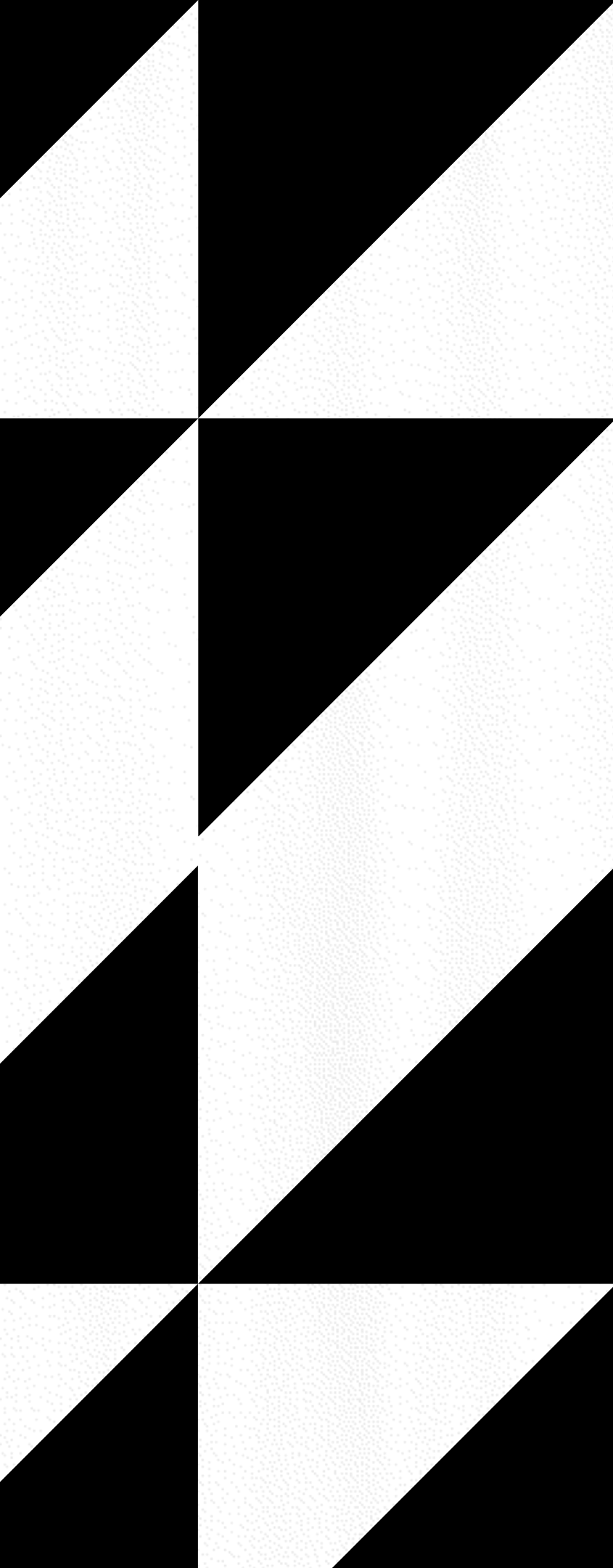
- Input A: 1010 (decimal 10)
- Input B: 0111 (decimal 7)
- Result: 0001 0001 (decimal 17, carry = 1)



These examples show how the carry bit signals when the sum exceeds 4 bits.

Example 2

- Input A: 1100 (decimal 12)
- Input B: 1010 (decimal 10)
- Result: 0110 0001 (decimal 22, carry = 1)



Installation Guide

- Download the file UnoArduSimV2.9.2.zip and extract it.
- Download base_trab_2.ino and save it in a folder (suggested name: Trabalho 2).
- Download ino_config.txt and myArduPrefs.txt and save them in the same folder.
- Open the Arduino simulator and load base_trab_2.ino.

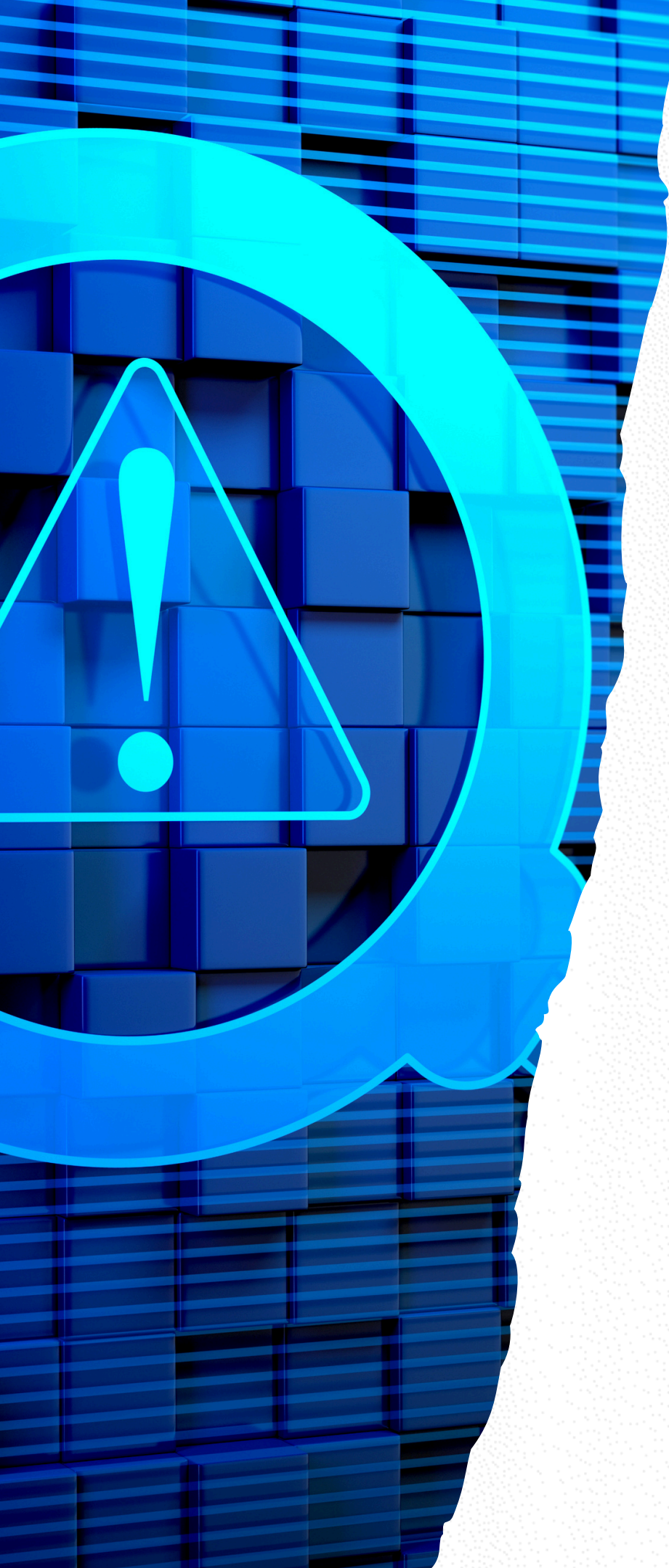
A large grid of binary digits (0s and 1s) arranged in a pattern that resembles a stylized letter 'A' or a similar abstract shape. The digits are black on a white background.

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FOR THE FUTURE

- Extend the project to support 8-bit or 16-bit numbers
- Replace LED outputs with 7-segment displays for better visualization
- Create a virtual simulation in Tinkercad or Proteus for testing
- Automate input testing to cover all possible binary combinations



Bugs and Fixes

- Input and output pins correctly configured
 - Sum and carry logic tested for all 4-bit input cases
 - Outputs correctly mapped to LEDs
 - No major bugs currently identified
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